UNIT 5 · STANDARD 6.G.1

Area of Trapezoids

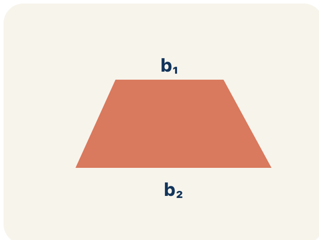
Lesson 5-2

Name: ______________________ Date: ______________________ Class: ______________________

Key Vocabulary

Level 1 support

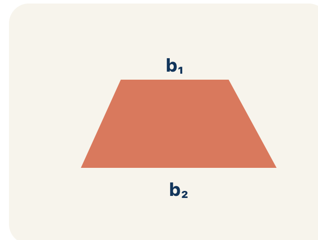
Picture first, then the word, then a plain-language meaning. Say each word out loud.



A table top wider than the bottom shelf — the top and bottom edges are parallel, the two sides slant inward

Trapezoid

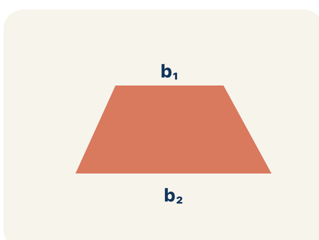
A four-sided shape with just one pair of parallel sides.



If the bottom of the trapezoid is 10 ft, then b1 = 10 ft in the formula $A = ½(b1 + b2) × h$

Base 1 (b1)

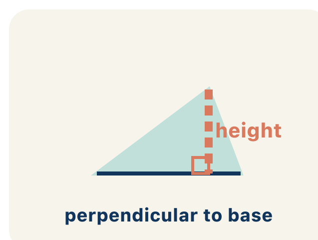
One of the two parallel sides of a trapezoid.



If the top of the trapezoid is 6 ft, then b2 = 6 ft; both bases are parallel to each other

Base 2 (b2)

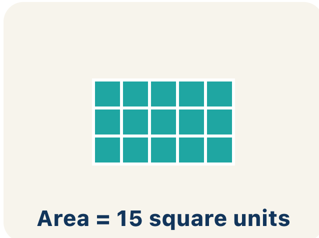
The other parallel side of a trapezoid.



A dashed vertical line from the top base straight down to the bottom base at a 90° angle — NOT the slanted side

Height

The straight-up distance between the two parallel sides.



A trapezoid with $b_1 = 10$, $b_2 = 6$, $h = 4$ has area = $\frac{1}{2}(10 + 6) \times 4 = 32$ sq units

Area

How much space is inside a flat shape.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can find the area of a trapezoid using the formula A = ½(b1 + b2) × h.

1️⃣ Notice

Your architecture firm is designing a trapezoidal window for a modern building.

2️⃣ Key idea

I can find the area of a trapezoid using the formula A = ½(b1 + b2) × h.

3️⃣ Words I'll use

Trapezoid **Base 1 (b1)** **Base 2 (b2)** **Height** **Area**

🤔 Think About It

- What shape is the window?
- How many parallel sides does it have?
- What measurements are given?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 I do — watch

Follow each step as your teacher solves it.

Problem: What is the area of a trapezoid with bases 6 cm and 10 cm, and height 4 cm?

A. 32 sq cm

B. 64 sq cm

C. 24 sq cm

D. 40 sq cm

Step 1 A = ½ × (b1 + b2) × h = ½ × (6 + 10) × 4 = ½ × 16 × 4 = 32 sq cm.

✅ **Answer:** A. 32 sq cm

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: A trapezoid has an area of 45 sq ft, bases of 7 ft and 11 ft. What is the height?

A. 5 ft

B. 9 ft

C. 3 ft

D. 10 ft

Step 1 ____________________

Step 2 ____________________

Answer: ____________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: What is the first step when finding the area of a trapezoid?

A. Add the two bases together

B. Multiply the two bases

C. Subtract the shorter base from the longer base

D. Divide the height by 2

Show your work:

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The trapezoidal window has a top edge of 4 feet and a bottom edge of 8 feet. Why does the area formula use BOTH bases instead of just one?

Level 1 support

Start here: We use both bases because the top and bottom are different lengths.

💡 **Need a hint?**

Sentence starters (optional)

We use both bases because the top and bottom are ____.

Usamos las dos bases porque la parte de arriba y la de abajo son ____.

If I used only one base, the area would be ____.

Si usara solo una base, el área sería ____.

Word bank: trapezoid | base 1 (b1) | base 2 (b2) | height | area

Listen for: *Students explain a trapezoid has two parallel sides of different lengths, so using only one would over- or under-count the area; strong answers mention averaging the bases.*

Level 2

The wonder prompt asks what happens if both bases are equal. Predict the shape and explain how the trapezoid formula would simplify in that case.

- If b1 equals b2, the shape becomes a ____ because ____.
- The formula simplifies to ____, since ____.

EXPLORE

On the grid you copied the trapezoid (bases 8 and 4, height 5), flipped it, and joined it to the original. What shape formed, and how does that explain the 1/2 in the formula?

Level 1 support

Start here: Two trapezoids make a parallelogram, so one trapezoid is half of it.

💡 **Need a hint?**

Sentence starters (optional)

When I flip and join the trapezoid, I make a ____.

Cuando volteo y uno el trapecio, formo un ____.

I added the two bases first because ____.

Primero sumé las dos bases porque ____.

The area is 1/2 times (8 plus 4) times 5, which equals ____.

El área es 1/2 por (8 más 4) por 5, que es ____.

Word bank: **trapezoid** **parallelogram** **base 1 (b1)** **base 2 (b2)** **height**

Listen for: *A strong answer names a parallelogram, links the 1/2 to taking half of two combined trapezoids, and computes area = 1/2 x 12 x 5 = 30 sq ft.*

Level 2

Compare finding a trapezoid's area to averaging the two bases and multiplying by the height. Are these the same method? Justify with the window numbers.

- Averaging the bases gives ____, and multiplying by the height gives ____, which ____ the formula answer.
- These methods are ____ because ____.

CONNECT

The builder pours concrete for a trapezoidal driveway with edges of 20 ft and 12 ft and a height of 15 ft; one bag covers 6 sq ft. Walk through finding the number of bags.

Level 1 support

Start here: Find the trapezoid area first, then divide by what one bag covers.

💡 **Need a hint?**

Sentence starters (optional)

The driveway area is 1/2 times (20 plus 12) times 15, which equals ____ square feet.

El área de la entrada es 1/2 por (20 más 12) por 15, que es ____ pies cuadrados.

He needs ____ bags because ____ divided by 6 equals ____.

Necesita ____ bolsas porque ____ entre 6 es ____.

Word bank: trapezoid | base 1 (b1) | base 2 (b2) | height | area

Listen for: *Students compute area = 240 sq ft, then 240 / 6 = 40 bags, and explain why dividing by coverage gives the bag count.*

Level 2

If the builder doubled only the height of the driveway, would the number of bags double? Justify using the formula.

- Doubling the height ____ the area because ____.
- So the number of bags would ____, since ____.

Write About the Math
The Writing Revolution

I can explain my steps using the words trapezoid, base, height, and area.

1. Kernel Sentence
subject + verb

Model: Trapezoid is a four-sided shape with just one pair of parallel sides.
Trapecio es una figura de cuatro lados con solo un par de lados paralelos.

Write a kernel sentence about trapezoid. Use a subject and a verb.
Escribe una oración base sobre trapecio. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion
because · but · so

Kernel: Trapezoid matters in math
Trapecio importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*
Trapezoid matters in math because ____.
Trapecio importa en matemáticas porque ____.

____

but *pero*
Trapezoid matters in math, but ____.
Trapecio importa en matemáticas, pero ____.

____

so *entonces*
Trapezoid matters in math, so ____.
Trapecio importa en matemáticas, entonces ____.

____

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about trapezoid.
Di un hecho verdadero sobre trapezoid.

Trapezoid ____.

Question
Pregunta

Ask a question about trapezoid.
Haz una pregunta sobre trapezoid.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about trapezoid.
Muestra entusiasmo sobre trapezoid.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with trapezoid.
Dile a un compañero qué hacer con trapezoid.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

We use both bases because the top and bottom are ____.
Usamos las dos bases porque la parte de arriba y la de abajo son ____.

If I used only one base, the area would be ____.
Si usara solo una base, el área sería ____.

When I flip and join the trapezoid, I make a ____.
Cuando volteo y uno el trapecio, formo un ____.

Try It

Solve on your own. Check the answer key when you are done.

1. A trapezoid has area 36 in² and bases 5 in and 7 in. What is its height?

A. 6 in

B. 3 in

C. 12 in

D. 18 in

Show your work:

2. Why is the trapezoid area formula ½(b1 + b2)h?

A. It averages the two bases, then multiplies by the height

B. It adds all four sides

C. It uses only the longer base

D. Trapezoids have no height

Show your work:

Stretch Your Thinking Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A trapezoid and a parallelogram both have a height of 8 ft. The trapezoid has bases of 5 ft and 11 ft. The parallelogram has a base of 8 ft. Which shape has the greater area? Explain your reasoning using the formulas.

Sentence starter: The trapezoid's area is ____ because ½ × (____ + ____) × ____ = ____. The parallelogram's area is ____ because ____ × ____ = ____. The ____ has a greater area by ____ sq ft.

Show your work:

Reflect — Exit Ticket

A trapezoid has bases of 9 inches and 5 inches, and a height of 6 inches. What is its area?

A. 42 sq in
B. 84 sq in
C. 27 sq in
D. 42 in

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. 5 ft *— $A = ½ \times (b_1 + b_2) \times h \rightarrow 45 = ½ \times 18 \times h \rightarrow 45 = 9h \rightarrow h = 5$ ft.*
2. **You Try (notes):** A. Add the two bases together *— The formula is $A = ½ \times (b_1 + b_2) \times h$. The first step is to add the two parallel bases.*
3. **Try It 1:** A. 6 in *— $36 = ½(5 + 7)h = 6h$, so $h = 6$ in.*
4. **Try It 2:** A. It averages the two bases, then multiplies by the height *— $½(b_1 + b_2)$ is the average of the two parallel sides, multiplied by the height.*
5. **Exit Ticket:** A. 42 sq in *— $A = 1/2 \times (b_1 + b_2) \times h = 1/2 \times (9 + 5) \times 6 = 1/2 \times 14 \times 6 = 42$ square inches.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Trapezoid is a four-sided shape with just one pair of parallel sides.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).